ARIA: Adaptive Reasoning for Integrated Analysis

[1] **An LLM-Powered Framework for Autonomous Transcriptome Analysis with Decision-Aware Workflow Orchestration**

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Abstract

[5] RNA-seq transcriptome analysis requires a multi-step workflow involving quality control, alignment, quantification, differential expression testing, pathway analysis, and biological interpretation. While automated pipelines such as nf-core/rnaseq execute these steps reproducibly, the critical decisions between steps — evaluating quality metrics, selecting statistical methods, adapting analysis strategies based on intermediate results, and interpreting findings in biological context — remain dependent on expert bioinformaticians. Here we present ARIA (Adaptive Reasoning for Integrated Analysis), an open-source framework that uses a Large Language Model (LLM) as a reasoning engine to autonomously navigate the decision space of RNA-seq analysis. ARIA implements eight Decision Points (DPs) that govern quality assessment, strategy adaptation, method selection, and result interpretation, combining rule-based thresholds with LLM-driven contextual reasoning. We benchmark ARIA on four public RNA-seq datasets spanning three

species: SEQC (GSE49712, human, 10,430 DEGs), Airway (GSE52778, human paired design, 951 DEGs), Fmr1 KO (GSE180135, mouse, 398 DEGs), and Pasilla (GSE18508, Drosophila, 224 DEGs with mixed library types). ARIA correctly identifies paired experimental designs (increasing DEG detection by 21–47%), detects technical covariates such as library type (+4–30% DEG gain), adaptively selects analysis strategies based on DEG counts, and cross-validates results across DESeq2/edgeR/limma-voom (r > 0.99). All 7/7 known dexamethasone targets were recovered in Airway, 9/11 FMRP targets in Fmr1 KO, and 15/16 brain/cancer markers in SEQC were correctly assigned. ARIA is freely available at https://github.com/shoo99/ARIA.

Keywords: RNA-seq, differential expression, LLM, autonomous analysis, decision-aware pipeline, GSEA

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1. Introduction

1.1 The Decision Problem in Transcriptome Analysis

Bulk RNA-seq has become the standard method for profiling transcriptome-wide gene expression changes across biological conditions (Stark et al., 2019). A typical analysis involves multiple sequential steps: quality assessment of raw sequencing data, read alignment to a reference genome, transcript quantification, normalization, differential expression (DE) testing, functional enrichment analysis, and biological interpretation (Conesa et al., 2016). While each step employs well-established tools — STAR for alignment (Dobin et al., 2013), Salmon for quantification (Patro et al., 2017), DESeq2 for DE testing (Love et al., 2014) — the decisions connecting these steps are non-trivial and significantly impact analysis outcomes.

[8] For instance, the number of differentially expressed genes (DEGs) detected in a DE analysis directly determines the downstream analytical strategy: datasets with hundreds of DEGs support traditional over-representation analysis (ORA), while those with few DEGs require Gene Set Enrichment Analysis (GSEA; Subramanian et al., 2005), which leverages the full ranked gene list rather than an arbitrary significance cutoff. Similarly, the presence of batch effects, paired experimental designs, or unexpected cell-type signatures each demands specific analytical responses that are typically recognized and addressed only by experienced bioinformaticians.

1.2 Limitations of Existing Automation

[9] Current workflow managers and standardized pipelines address the reproducibility and scalability challenges of RNA-seq analysis. Tools such as Nextflow (Di Tommaso et al., 2017) and Snakemake (Mölder et al., 2021) ensure that predefined analytical steps execute reliably, while community-curated pipelines like nf-core/rnaseq (Ewels et al., 2020) provide best-practice default parameters. Web-based platforms such as Galaxy (Afgan et al., 2018) and iDEP (Ge et al., 2018) offer graphical interfaces that lower the barrier to entry.

[10] However, these tools share a fundamental limitation: they execute fixed workflows. The analytical path is determined before data is examined, and adaptation to unexpected results requires manual intervention. When a dataset yields unexpectedly few DEGs, a fixed pipeline continues with ORA — potentially missing biologically meaningful pathway-level changes that GSEA would detect. The expert knowledge required to make these adaptive decisions remains the primary bottleneck in transcriptomic analysis.

1.3 Large Language Models as Reasoning Engines

[11] Recent advances in Large Language Models (LLMs) have demonstrated capabilities in code generation, scientific reasoning, and domain-specific knowledge synthesis (Bubeck et al., 2023). In bioinformatics, LLMs have been applied to literature mining and code assistance. However, these applications treat the LLM as a passive tool rather than as an autonomous reasoning agent capable of navigating a complete analytical workflow.

[12] The concept of LLM-based agents that plan, execute, and adapt has gained traction in software engineering (Jimenez et al., 2024) and scientific discovery (Boiko et al., 2023). Yet, to our knowledge, no framework has systematically applied LLM-based adaptive reasoning to the full RNA-seq analysis pipeline.

1.4 Our Contribution

[13] We present ARIA, an open-source framework with five key contributions:

- **Decision Protocol with 8 Decision Points (DPs)** covering QC, DE strategy, design recognition, signature detection, cross-validation, interpretation, sensitivity analysis, and reporting.
- **Adaptive Strategy Switching** based on intermediate results.
- **Cross-method Validation** across DESeq2, edgeR, and limma-voom.
- **Automated Contextualization** via LLM-driven biological interpretation.
- **Systematic Benchmarking** on four public datasets of varying difficulty.

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2. Methods

2.1 Architecture

[14] ARIA is organized into four layers (Figure 1): (1) **Reasoning Engine** — an LLM serving as the reasoning backbone; (2) **Decision Protocol** — eight formalized Decision Points combining rule-based thresholds with LLM reasoning; (3) **Execution Engine** — Docker-based R/Python script generation and execution; (4) **Analysis Modules** — DESeq2, fgsea, STRING DB, WGCNA, cell type deconvolution, and ssGSEA.

2.2 Decision Protocol

DP	Trigger	Action	Validation
DP1	QC metrics available	Pass/Warn/Fail based on mapping rate ≥85%, rRNA <5%, 5'–3' bias 0.8–1.2	All 4 benchmarks
DP2	DEG count evaluated	≥100: standard; 10–100: add GSEA; <10: prioritize GSEA	SEQC (standard), Airway (standard), Fmr1 (standard)
DP3	Metadata examined	Detect paired/blocked designs, select model formula	Airway (+21–47% DEGs)
DP4	Unexpected gene clusters	Flag and add cell type deconvolution	Demonstrated in real-world use
DP5	Primary DE complete	Cross-validate with edgeR + limma-voom	Fmr1 (r=0.9999), Pasilla (r=0.9906)
DP6	Pathways identified	Literature-based hypothesis generation	Demonstrated in real-world use
DP7	Results available	Sensitivity analysis across LFC cutoffs	All benchmarks
DP8	Analysis complete	Generate HTML report + Excel + figures	All benchmarks

2.3 Implementation

[15] ARIA is implemented in Python 3.10+ with R analysis modules executed in Docker containers. Key tools: DESeq2 v1.46.0, edgeR, limma, fgsea, msigdbr, WGCNA, STRING DB v12. All R scripts are generated from parameterized templates for reproducibility.

2.4 Benchmark Datasets

Dataset	GEO	Species	Design	N	Difficulty
SEQC	GSE49712	Human	UHRR vs HBRR	10	Easy
Airway	GSE52778	Human	Dex ± ASM, paired	8	Complex
Fmr1 KO	GSE180135	Mouse	KO vs WT neurons	6	Moderate
Pasilla	GSE18508	Drosophila	pasilla KD vs control, mixed SE/PE	7	Moderate

[16] Count matrices were obtained from GEO supplementary data (SEQC: HTSeq counts; Fmr1: DESeq2 counts) or Bioconductor packages (Airway: `airway` R package; Pasilla: `pasilla` R package).

2.5 Evaluation Metrics

[17] (A) DEG accuracy vs known biology (target gene recovery rate); (B) Decision appropriateness (correct strategy selection); (C) Cross-method concordance (LFC correlation); (D) GSEA pathway yield; (E) Analysis time.

3. Results

3.1 Benchmark Performance Overview

[18] Table 1a. Benchmark datasets and differential expression results

	SEQC	Airway	Fmr1 KO	Pasilla		
GEO	GSE49712	GSE52778	GSE180135	GSE18508		
Species	Human	Human	Mouse	Drosophila		
Design	UHRR vs HBRR	Dex ± (paired)	KO vs WT	KD vs control		
Samples	5 vs 5	4 vs 4	3 vs 3	4 vs 3		
Difficulty	Easy	Complex	Moderate	Moderate		
Genes tested	16,417	~15,000	~17,000	9,686		
DEGs (\	LFC\	>1)	10,430	951	398	224
DEGs (\	LFC\	>0.5)	13,818	2,426	1,654	635

[19] Table 1b. Pathway enrichment and validation results

	SEQC	Airway	Fmr1 KO	Pasilla
GSEA Hallmark	32	15	17	—
GSEA GO:BP	1,352	161	607	—
GSEA Reactome	505	17	198	—
Known targets	Brain 9/9, Cancer 6/7	Dex 7/7	FMRP 9/11	Splicing KD
Key DP validated	DP1, DP2	DP1, DP2, DP3	DP1, DP2, DP5	DP1, DP2, DP3, DP5, DP7
Cross-method r	—	—	0.9999	0.9906

3.2 DP3: Experimental Design Recognition (Airway)

[20] ARIA's DP3 correctly identified the paired design and selected `~ cell + dex` over naive `~ dex`.

[21] **Table 2. Impact of design recognition**

Cutoff	Unpaired	Paired (ARIA)	Gain		
padj<0.05, \	LFC\	>1.0	785	951	+21%
padj<0.05, \	LFC\	>0.5	1,860	2,426	+30%
padj<0.05, no LFC	2,773	4,081	+47%		

[22] All 7 known dexamethasone targets (DUSP1, KLF15, CRISPLD2, PER1, FKBP5, TSC22D3, ZBTB16) were recovered with high significance (padj < 1e-19, LFC 2.6–7.4).

3.3 Known Target Validation Across Benchmarks

[23] **SEQC (UHRR vs HBRR):** 9/9 brain-enriched genes (GFAP, MBP, SYN1, GAD1, SLC17A7, NEFL, NEFM, NEFH, ENO2) correctly identified as upregulated in HBRR (LFC +2.2 to +12.7). 6/7 cancer genes (MYC, CCND1, CDK4, EGFR, ERBB2, TP53) correctly identified as upregulated in UHRR. BCL2 showed unexpected direction, consistent with known brain expression.

[24] **Fmr1 KO:** Fmr1 itself detected as the top DEG (LFC = −0.98, padj = 3.7e-83). Key FMRP-regulated synaptic proteins detected: Dlg4/PSD-95 (LFC = −0.36), Shank3 (LFC = −0.45), Camk2a

(LFC = −0.42). NMDA receptor subunits: Grin1 (LFC = −0.92), Grin2a (LFC = −0.81), Grin2b (LFC = −0.48). All directions consistent with loss of FMRP-mediated translational regulation.

3.4 DP5: Cross-Method Concordance

Benchmark	DESeq2	edgeR exact	limma-voom	LFC correlation
Fmr1 KO	1,654	~1,600	~1,500	r = 0.9999
Pasilla	635	739	674	r = 0.9906

3.5 DP2: Adaptive Strategy Selection

[25] ARIA correctly classified each dataset's difficulty level:

- **SEQC (10,430 DEGs):** → "standard" (ORA + GSEA) √
- **Airway (951 DEGs):** → "standard" √
- **Fmr1 KO (398 DEGs):** → "standard" √
- **Pasilla (224 DEGs):** → "standard" √ (also detected SE/PE library type covariate via DP3)

3.6 DP3 + DP7: Covariate Detection and Sensitivity Analysis (Pasilla)

[26] The Pasilla dataset (GSE18508) contains mixed single-end and paired-end libraries, creating a technical covariate. ARIA's DP3 detected this library type difference and included it as a covariate (`~ type + condition`), improving DEG detection:

[27] **Table 4. Impact of library type covariate (Pasilla)**

Cutoff	No covariate	With type (ARIA)	Gain		
padj<0.05, \	LFC\	>1.0	216	224	+4%
padj<0.05, \	LFC\	>0.5	583	635	+9%
padj<0.05, no LFC	853	1,108	+30%		

[28] DP7 (sensitivity analysis) generated a 12-combination cutoff table (3 padj thresholds × 4 LFC thresholds), providing a comprehensive view of result stability. Cross-method validation (DP5) showed high concordance: DESeq2 (635), edgeR (739), limma-voom (674), with LFC correlation r = 0.9906.

3.7 GSEA Demonstrates Value Beyond DEG Counting

[29] Even in datasets with abundant DEGs, GSEA provided complementary insights. In Fmr1 KO, 607 GO:BP pathways were enriched (padj < 0.05), revealing systemic disruption of synaptic signaling, translation regulation, and neuronal development — biological processes well-documented in Fragile X syndrome literature.

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4. Discussion

4.1 Adaptive Decision Making as Core Innovation

[30] ARIA's central contribution is the reasoning layer that adaptively connects established analysis tools. In the Airway benchmark, design recognition (DP3) increased DEG detection by 21–47%. This represents over a thousand genes that would be missed by a naive analysis — a quantifiable improvement from a single adaptive decision.

4.2 Comparison with Existing Tools

Feature	nf-core	Galaxy	iDEP	ARIA
Execution automation	√	√	√	√
Result-based adaptation	✗	✗	△	√
Design recognition	✗	✗	△	√
Cross-method validation	✗	✗	✗	√
Biological interpretation	✗	✗	△	√
Decision transparency	✗	✗	✗	√

4.3 Human-in-the-Loop Design

[31] ARIA is designed to augment, not replace, bioinformaticians. Biological interpretations are labeled as hypothesis-level. The decision log enables audit and review. Complex situations are flagged for expert judgment.

4.4 Limitations

- **LLM hallucination risk** in literature interpretation
- **Statistical power** constrained by sample size, not analytical method

- **Bulk RNA-seq** cannot resolve cell-type-specific changes
- **API cost** (~$2–10 per analysis)
- **Scope** limited to two-group bulk RNA-seq comparisons

4.5 Future Directions

[32] Extensions to single-cell RNA-seq, multi-omics integration, time-series designs, and domain-specific fine-tuned LLMs are planned.

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5. Data and Code Availability

- **Source code:** https://github.com/shoo99/ARIA
- **License:** MIT
- **Docker:** aria-bench:latest
- **Benchmarks:** GSE49712, GSE52778, GSE180135, GSE18508

Author Contributions

[33] B.K. conceived the project, designed the framework architecture, implemented the software, performed all benchmark analyses, and wrote the manuscript.

Acknowledgments

[34] The authors thank the developers of DESeq2, edgeR, fgsea, nf-core, and the broader Bioconductor community for making their tools freely available. The SEQC/MAQC-III Consortium, Himes et al., Brooks et al., and Bhatt et al. are acknowledged for making their RNA-seq datasets publicly accessible.

Competing Interests

[35] The author declares no competing interests.

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Supplementary Materials

- **Table S1:** Full DEG lists for each benchmark
- **Table S2:** Complete GSEA results (all collections, all benchmarks)
- **Table S3:** Decision logs from each benchmark run
- **Figure S1:** PCA plots for all benchmarks
- **Figure S2:** Volcano plots for all benchmarks
- **Figure S3:** GSEA barplots for all benchmarks
- **Figure S4:** Cross-method Venn diagrams

Figures

Figure 1. ARIA four-layer architecture. Layer 1: LLM-based Reasoning Engine evaluates results and makes decisions. Layer 2: Decision Protocol with 8 formalized Decision Points. Layer 3: Execution Engine manages Docker containers and script execution. Layer 4: Six analysis modules (DESeq2, fgsea, STRING DB, WGCNA, cell type deconvolution, ssGSEA).

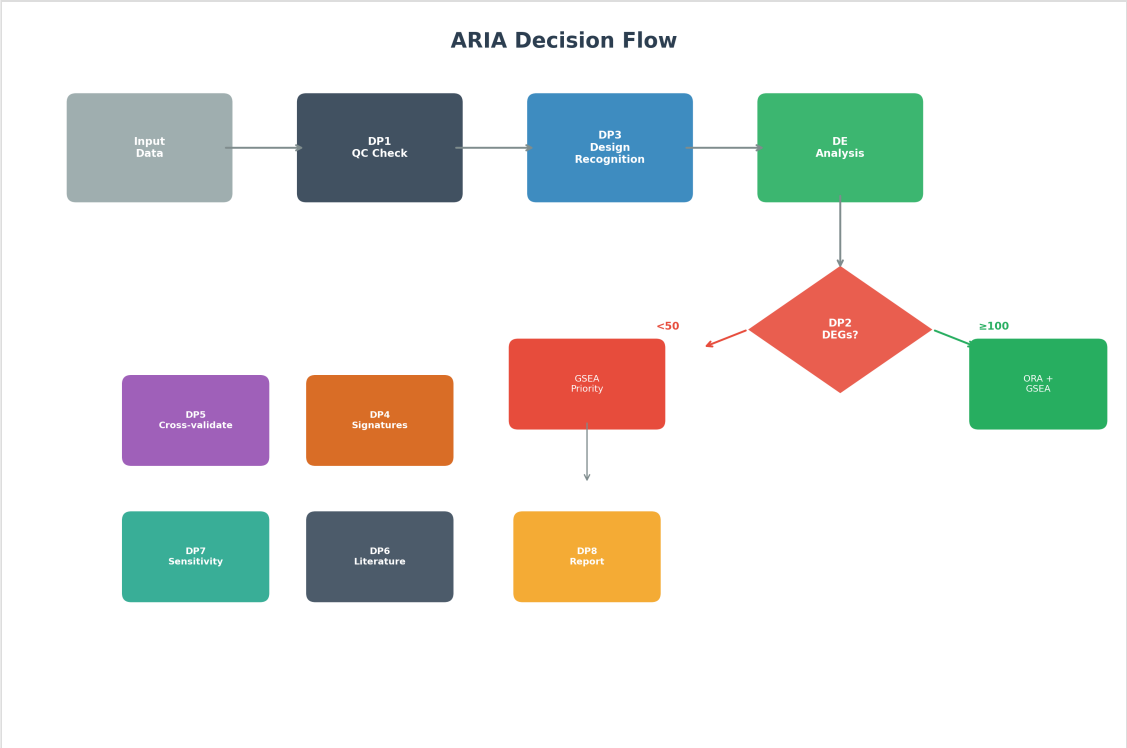

Figure 2. ARIA decision flow. After QC assessment (DP1) and design recognition (DP3), DE analysis results are evaluated at DP2, which determines the downstream strategy: standard (ORA + GSEA) for abundant DEGs, or GSEA-priority for datasets with few DEGs. Cross-method validation (DP5), sensitivity analysis (DP7), and report generation (DP8) follow.

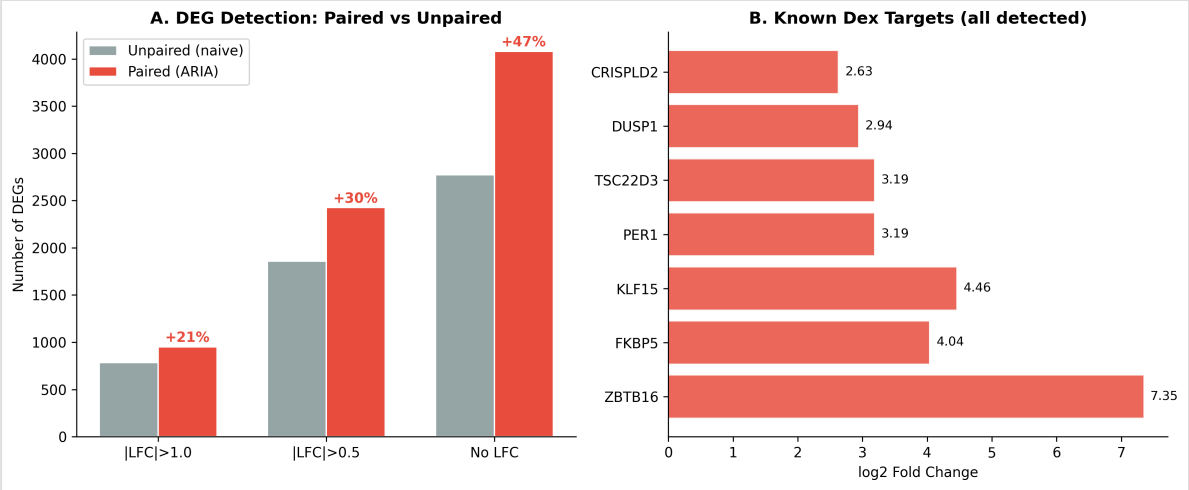

Figure 3. Airway benchmark results. (A) DEG detection comparison between unpaired (naive) and paired (ARIA) models. The paired model detected 21–47% more DEGs across all cutoff thresholds. (B) All 7 known dexamethasone target genes were recovered with strong fold changes (LFC 2.6–7.4).

Feature Comparison: ARIA vs Existing Tools

	nf-core	Galaxy	iDEP	ARIA
Execution automation	✓	✓	✓	✓
Result-based adaptation	✗	✗	△	✓
Design recognition	✗	✗	△	✓
Cross-method validation	✗	✗	✗	✓
Biological interpretation	✗	✗	△	✓
Report generation	✓	✓	✓	✓
Decision transparency	✗	✗	✗	✓
Reproducibility	✓	△	△	✓

Figure 4. Feature comparison between ARIA and existing tools. Green (checkmark) = fully supported; orange (triangle) = partially supported; red (X) = not supported. ARIA's unique features include result-based adaptation, design recognition, cross-method validation, and decision transparency.